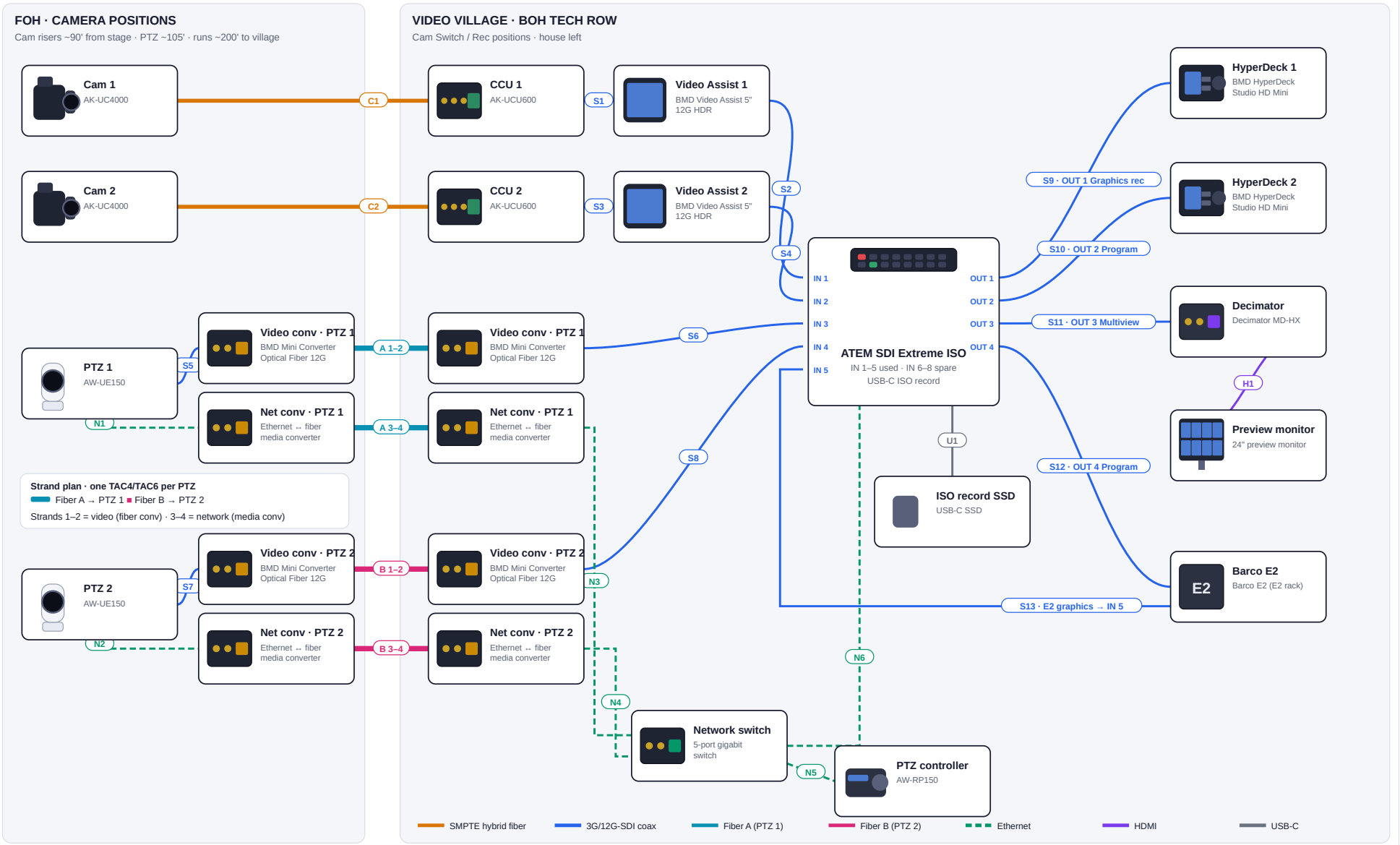
Camera & Record Wiring

digerati



Formats: the ATEM SDI Extreme ISO is HD only (3G-SDI). Set both CCUs and both PTZs to **1080p** at the show frame rate before patching. **Outputs** are assigned in ATEM Software Control → Settings → Outputs. **PTZ power** is local AC at the camera position; fiber doesn't carry PoE. **Fiber A** (PTZ 1) and **Fiber B** (PTZ 2) are separate TAC4/TAC6 runs: strands 1–2 video, 3–4 network.

ATEM ASSIGNMENTS

INPUT	SOURCE	PATH
SDI IN 1	Cam 1 · AK-UC4000	CCU 1 → Video Assist 1 loop
SDI IN 2	Cam 2 · AK-UC4000	CCU 2 → Video Assist 2 loop
SDI IN 3	PTZ 1 · AW-UE150	Fiber A strands 1–2 via Mini Converter 12G
SDI IN 4	PTZ 2 · AW-UE150	Fiber B strands 1–2 via Mini Converter 12G
SDI IN 5	E2 graphics	From the E2 rack, 25' SDI
SDI IN 6–8	Spare	

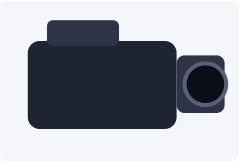
OUTPUT	FEED	GOES TO
SDI OUT 1	Graphics record	HyperDeck Studio HD Mini #1
SDI OUT 2	Program	HyperDeck Studio HD Mini #2
SDI OUT 3	Multiview	Decimator MD-HX → HDMI → 24" preview monitor
SDI OUT 4	Program	Barco E2 input (E2 station)

CABLE RUNS · PORT TO PORT

ID	FROM		TO	CABLE	LENGTH	NOTES
C1	Cam 1 Optical (LEMO)	→	CCU 1 Optical (LEMO)	SMPTE hybrid fiber (camera cable)	~200'	Camera riser → cable pick → DS truss → 30' run → down to video village
C2	Cam 2 Optical (LEMO)	→	CCU 2 Optical (LEMO)	SMPTE hybrid fiber (camera cable)	~200'	Same path as C1
S1	CCU 1 12G-SDI OUT 1	→	Video Assist 1 SDI IN	3G/12G-SDI coax (BNC)	6'	
S2	Video Assist 1 SDI OUT (loop)	→	ATEM SDI IN 1	3G/12G-SDI coax (BNC)	6'	
S3	CCU 2 12G-SDI OUT 1	→	Video Assist 2 SDI IN	3G/12G-SDI coax (BNC)	6'	
S4	Video Assist 2 SDI OUT (loop)	→	ATEM SDI IN 2	3G/12G-SDI coax (BNC)	6'	
S5	PTZ 1 3G-SDI OUT	→	Video conv · PTZ 1 (cam end) SDI IN	3G/12G-SDI coax (BNC)	3'	At the PTZ position
A1-2	Video conv · PTZ 1 (cam end) Optical (LC)	→	Video conv · PTZ 1 (village) Optical (LC)	Fiber A · TAC4/6 to PTZ 1	~200'	Fiber A, strands 1–2 · PTZ 1 video
S6	Video conv · PTZ 1 (village) SDI OUT	→	ATEM SDI IN 3	3G/12G-SDI coax (BNC)	6'	
N1	PTZ 1 LAN	→	Net conv · PTZ 1 (cam end) LAN	Ethernet (Cat5e/6)	6'	Power PTZ 1 with its AC adapter (no PoE over fiber)
A3-4	Net conv · PTZ 1 (cam end) Fiber (LC)	→	Net conv · PTZ 1 (village) Fiber (LC)	Fiber A · TAC4/6 to PTZ 1	~200'	Fiber A, strands 3–4 · PTZ 1 network
N3	Net conv · PTZ 1 (village) LAN	→	Network switch Port 1	Ethernet (Cat5e/6)	6'	
S7	PTZ 2 3G-SDI OUT	→	Video conv · PTZ 2 (cam end) SDI IN	3G/12G-SDI coax (BNC)	3'	At the PTZ position
B1-2	Video conv · PTZ 2 (cam end) Optical (LC)	→	Video conv · PTZ 2 (village) Optical (LC)	Fiber B · TAC4/6 to PTZ 2	~200'	Fiber B, strands 1–2 · PTZ 2 video

ID	FROM		TO	CABLE	LENGTH	NOTES
S8	Video conv · PTZ 2 (village) SDI OUT	→	ATEM SDI IN 4	3G/12G-SDI coax (BNC)	6'	
N2	PTZ 2 LAN	→	Net conv · PTZ 2 (cam end) LAN	Ethernet (Cat5e/6)	6'	Power PTZ 2 with its AC adapter
B3-4	Net conv · PTZ 2 (cam end) Fiber (LC)	→	Net conv · PTZ 2 (village) Fiber (LC)	Fiber B · TAC4/6 to PTZ 2	~200'	Fiber B, strands 3–4 · PTZ 2 network
N4	Net conv · PTZ 2 (village) LAN	→	Network switch Port 2	Ethernet (Cat5e/6)	6'	
N5	Network switch Port 3	→	PTZ controller LAN	Ethernet (Cat5e/6)	6'	
N6	Network switch Port 4	→	ATEM Ethernet	Ethernet (Cat5e/6)	6'	ATEM Software Control laptop on port 5
S13	Barco E2 Graphics output (assign)	→	ATEM SDI IN 5	3G/12G-SDI coax (BNC)	25'	E2 graphics into the switcher
S9	ATEM SDI OUT 1 · Graphics record	→	HyperDeck 1 SDI IN	3G/12G-SDI coax (BNC)	6'	
S10	ATEM SDI OUT 2 · Program	→	HyperDeck 2 SDI IN	3G/12G-SDI coax (BNC)	6'	
S11	ATEM SDI OUT 3 · Multiview	→	Decimator SDI IN	3G/12G-SDI coax (BNC)	6'	
H1	Decimator HDMI OUT	→	Preview monitor HDMI IN	HDMI	6'	
S12	ATEM SDI OUT 4 · Program	→	Barco E2 SDI input (assign)	3G/12G-SDI coax (BNC)	25'	Program to the E2 rack
U1	ATEM USB-C	→	ISO SSD USB-C	USB-C	—	ISO recording of all inputs

WHAT EACH PIECE DOES

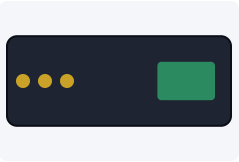


Panasonic AK-UC4000 (×2)

4K studio/field camera. Video, tally, intercom and power all ride one SMPTE hybrid fiber cable to its CCU.

Optical (LEMO/Tajimi) → CCU

Product page & photos ↗

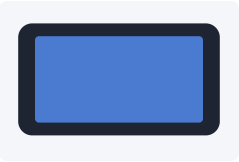


Panasonic AK-UCU600 (×2)

Camera control unit. Powers the camera over the fiber and outputs the picture as SDI. Shade from the RCP if it's in the kit.

SMPTE fiber in · 12G-SDI OUT 1 → Video Assist. **Set SDI output to 1080p HD**; the ATEM is HD-only.

Product page & photos ↗



Blackmagic Video Assist 5" 12G HDR (×2)

5" monitor and recorder inline on each studio camera. Records a camera ISO to SD and loops the signal on to the switcher.

SDI IN ← CCU · SDI OUT (loop) → ATEM IN 1/2

Product page & photos ↗

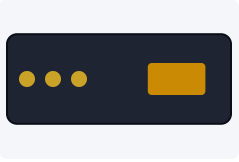


Panasonic AW-UE150 (×2)

4K PTZ camera, controlled from the AW-RP150 over the network.

3G-SDI OUT → Fiber A · LAN → media converter · **local AC power** (no PoE over fiber)

Product page & photos ↗

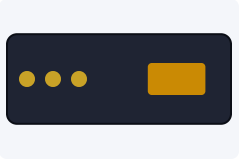


Blackmagic Mini Converter Optical Fiber 12G (×4)

SDI ↔ fiber converter. One at each PTZ and one in video village, linked by strands 1–2 of that camera's fiber (A or B).

SDI IN ← PTZ · strands 1–2 · SDI OUT → ATEM IN 3/4

Product page & photos ↗

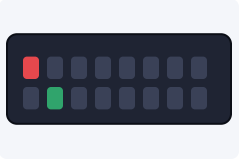


Ethernet ↔ fiber media converter (×4)

A pair per PTZ, one at the camera and one in the village, on strands 3–4 of that camera's fiber.

PTZ LAN → converter · strands 3–4 → converter → switch

Generic, any brand

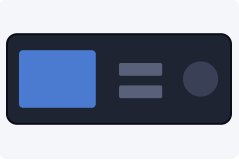


Blackmagic ATEM SDI Extreme ISO

8-input HD SDI switcher. Records ISOs of every input plus program to USB-C. Four SDI outputs, each assignable in ATEM Software Control.

IN 1–4 cameras · IN 5 E2 graphics · OUT 1 Graphics rec · OUT 2 Program · OUT 3 Multiview · OUT 4 Program → E2

Product page & photos ↗

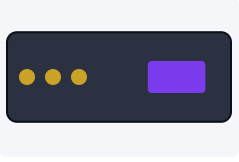


Blackmagic HyperDeck Studio HD Mini (×2)

Compact SDI recorder, about the size of a Web Presenter. Records to SD cards.

HyperDeck 1 ← OUT 1 (graphics record) · HyperDeck 2 ← OUT 2 (program)

Product page & photos ↗

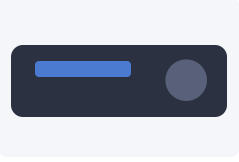


Decimator MD-HX

Bidirectional SDI ↔ HDMI converter with scaling. Turns the ATEM multiview SDI into HDMI for the preview monitor.

SDI IN ← ATEM OUT 3 · HDMI OUT → 24" monitor

Product page & photos ↗



Panasonic AW-RP150

PTZ remote controller. Joystick, zoom, presets for both PTZs.

LAN → switch (same network as the PTZs)

Product page & photos ↗

CABLE & GEAR PULL

ITEM	QTY	NOTES
SMPTE hybrid fiber camera cable	**2**	~250' each; confirm lengths in the camera kit
TAC4 or TAC6 single-mode tactical fiber · Fiber A	**1**	~250' to PTZ 1. Strands 1–2 video, 3–4 network (5–6 spare on TAC6)
TAC4 or TAC6 single-mode tactical fiber · Fiber B	**1**	~250' to PTZ 2. Same strand plan
LC–LC duplex patch cords	**8**	Fan-out to each converter, 4 at each end (if the tactical isn't LC-terminated)
Blackmagic Mini Converter Optical Fiber 12G	**4**	2 at the PTZs, 2 in the village
Ethernet ↔ fiber media converter	**4**	1 pair per PTZ
BNC 3G-SDI, 3'	**2 + 1 spare**	S5, S7 at the PTZs
BNC 3G-SDI, 6'	**9 + 2 spare**	S1–S4, S6, S8–S11 in the village
BNC 3G-SDI, 25'	**2**	S12 program to E2, S13 E2 graphics to ATEM IN 5
Ethernet Cat6, 6'	**7**	N1–N6 + ATEM control laptop
HDMI, 6'	**1**	H1 Decimator → preview monitor
USB-C SSD (1–2 TB)	**1**	ATEM ISO record. The gear list has the 2 TB drive at qty 0
SD cards	**per recorder**	Video Assists + HyperDecks, formatted before doors
AC power at the PTZ position	**2 circuits**	PTZ power + converters (no PoE over fiber)

> Runs to the FOH camera positions are estimated from the CAD: camera riser → cable pick → DS truss → 30' truss run → down to video village, about 200'. Pull 250'.

CABLE LABELS · PRINT TWO OF EACH, ONE PER END

C1
Cam 1 · Optical (LEMO)
→ CCU 1 · Optical (LEMO)

C1
Cam 1 · Optical (LEMO)
→ CCU 1 · Optical (LEMO)

C2
Cam 2 · Optical (LEMO)
→ CCU 2 · Optical (LEMO)

C2
Cam 2 · Optical (LEMO)
→ CCU 2 · Optical (LEMO)

S1
CCU 1 · 12G-SDI OUT 1
→ Video Assist 1 · SDI IN

S1
CCU 1 · 12G-SDI OUT 1
→ Video Assist 1 · SDI IN

S2
Video Assist 1 · SDI OUT (loop)
→ ATEM · SDI IN 1

S2
Video Assist 1 · SDI OUT (loop)
→ ATEM · SDI IN 1

S3
CCU 2 · 12G-SDI OUT 1
→ Video Assist 2 · SDI IN

S3
CCU 2 · 12G-SDI OUT 1
→ Video Assist 2 · SDI IN

S4
Video Assist 2 · SDI OUT (loop)
→ ATEM · SDI IN 2

S4
Video Assist 2 · SDI OUT (loop)
→ ATEM · SDI IN 2

S5
PTZ 1 · 3G-SDI OUT
→ Video conv · PTZ 1 (cam end) · SDI IN

S5
PTZ 1 · 3G-SDI OUT
→ Video conv · PTZ 1 (cam end) · SDI IN

A1-2
Video conv · PTZ 1 (cam end) · Optical (LC)
→ Video conv · PTZ 1 (village) · Optical (LC)

A1-2
Video conv · PTZ 1 (cam end) · Optical (LC)
→ Video conv · PTZ 1 (village) · Optical (LC)

S6
Video conv · PTZ 1 (village) · SDI OUT
→ ATEM · SDI IN 3

S6
Video conv · PTZ 1 (village) · SDI OUT
→ ATEM · SDI IN 3

N1
PTZ 1 · LAN
→ Net conv · PTZ 1 (cam end) · LAN

N1
PTZ 1 · LAN
→ Net conv · PTZ 1 (cam end) · LAN

A3-4
Net conv · PTZ 1 (cam end) · Fiber (LC)
→ Net conv · PTZ 1 (village) · Fiber (LC)

A3-4
Net conv · PTZ 1 (cam end) · Fiber (LC)
→ Net conv · PTZ 1 (village) · Fiber (LC)

N3
Net conv · PTZ 1 (village) · LAN
→ Network switch · Port 1

N3
Net conv · PTZ 1 (village) · LAN
→ Network switch · Port 1

S7
PTZ 2 · 3G-SDI OUT
→ Video conv · PTZ 2 (cam end) · SDI IN

S7
PTZ 2 · 3G-SDI OUT
→ Video conv · PTZ 2 (cam end) · SDI IN

B1-2
Video conv · PTZ 2 (cam end) · Optical (LC)
→ Video conv · PTZ 2 (village) · Optical (LC)

B1-2
Video conv · PTZ 2 (cam end) · Optical (LC)
→ Video conv · PTZ 2 (village) · Optical (LC)

S8
Video conv · PTZ 2 (village) · SDI OUT
→ ATEM · SDI IN 4

S8
Video conv · PTZ 2 (village) · SDI OUT
→ ATEM · SDI IN 4

N2
PTZ 2 · LAN
→ Net conv · PTZ 2 (cam end) · LAN

N2
PTZ 2 · LAN
→ Net conv · PTZ 2 (cam end) · LAN

B3-4
Net conv · PTZ 2 (cam end) · Fiber (LC)
→ Net conv · PTZ 2 (village) · Fiber (LC)

B3-4
Net conv · PTZ 2 (cam end) · Fiber (LC)
→ Net conv · PTZ 2 (village) · Fiber (LC)

N4
Net conv · PTZ 2 (village) · LAN
→ Network switch · Port 2

N4
Net conv · PTZ 2 (village) · LAN
→ Network switch · Port 2

N5
Network switch · Port 3
→ PTZ controller · LAN

N5
Network switch · Port 3
→ PTZ controller · LAN

N6
Network switch · Port 4
→ ATEM · Ethernet

N6
Network switch · Port 4
→ ATEM · Ethernet

S13
Barco E2 · Graphics output (assign)
→ ATEM · SDI IN 5

S13
Barco E2 · Graphics output (assign)
→ ATEM · SDI IN 5

S9
ATEM · SDI OUT 1 · Graphics record
→ HyperDeck 1 · SDI IN

S9
ATEM · SDI OUT 1 · Graphics record
→ HyperDeck 1 · SDI IN

S10
ATEM · SDI OUT 2 · Program
→ HyperDeck 2 · SDI IN

S10
ATEM · SDI OUT 2 · Program
→ HyperDeck 2 · SDI IN

S11
ATEM · SDI OUT 3 · Multiview
→ Decimator · SDI IN

S11
ATEM · SDI OUT 3 · Multiview
→ Decimator · SDI IN

H1
Decimator · HDMI OUT
→ Preview monitor · HDMI IN

H1
Decimator · HDMI OUT
→ Preview monitor · HDMI IN

S12
ATEM · SDI OUT 4 · Program
→ Barco E2 · SDI input (assign)

S12
ATEM · SDI OUT 4 · Program
→ Barco E2 · SDI input (assign)

U1
ATEM · USB-C
→ ISO SSD · USB-C

U1
ATEM · USB-C
→ ISO SSD · USB-C